AACENTURE ROUND 1 TECHNICAL QUESTIONS

[Q001]. Determine which of the following are VALID identifiers. If invalid, state the reason.

(a) sample1 (b) 5sample (c) data_7 (d) return (e) #fine

(f) variable (g) 91-080-100 (h) name & age (i) _val (j) name_and_age

Ans. (a) VALID

(b) Invalid, since an identifier must begin with a letter or an underscore

(c) VALID

(d) Invalid, since return is a reserved word

(e) Invalid, since an identifier must begin with a letter or an underscore

(f) VALID

(g) Invalid, since an identifier must begin with a letter or an underscore

(h) Invalid, since blank spaces are not allowed

(i) VALID

(j) VALID

[Q002]. Determine which of the following are VALID character constants. If invalid, state the reason.

(a) 'y' (b) '\r' (c) 'Y' (d) '@' (e) '/r'

(f) 'word' (g) '\0' (h) '\?' (i) '\065' (j) '\'' (k) ' '

Ans. (a) VALID

(b) VALID

(c) VALID

(d) VALID

(e) Invalid, since escape sequences must be written with a backward slash (i.e. \)

(f) Invalid, since a character constant cannot consist of multiple characters

(g) VALID (null-character escape sequence)

(h) VALID

(i) VALID (Octal escape sequence)

(j) VALID

(k) VALID

[Q003]. Determine which of the following are VALID string constants. If invalid, state the reason.

(a) 'Hi Friends' (b) "abc,def,ghi" (c) "Qualification

(d) "4325.76e-8" (e) "Don\'t sleep" (f) "He said, "You\'re great"

(g) "" (h) " " (i) "Rs.100/-"

Ans. (a) Invalid, since a string constant must be enclosed in double quotation marks

(b) VALID

(c) Invalid, since trailing quotation mark is missing

(d) VALID

(e) VALID (single-quote escape sequence)

(f) Invalid, since the quotation marks and (optionally) apostrophe within the string cannot be expressed without the escape sequences.

(g) VALID

(h) VALID

(i) VALID

[Q004]. Determine which of the following numerical values are valid constants. If a constant is

valid, specify whether it is integer or real. Also, specify the base for each valid integer constant.

(a) 10,500 (b) 080 (c) 0.007 (d) 5.6e7 (e) 5.6e-7

(f) 0.2e-0.3 (g) 0.2e 0.3 (h) 0xaf9s82 (i) 0XABCDEFL (j) 0369CF

(k) 87654321l (l) 87654321

Ans. (a) Invalid, since illegal character(,)

(b) VALID

(c) VALID

(d) VALID

(e) VALID

(f) VALID

(g) Invalid, since illegal character(blank space)

(h) Invalid, since illegal character(s)

(i) VALID

(j) Invalid, since illegal characters (9, C, F), if intended as an octal constant.

(k) VALID

(l) VALID

[Q005]. Determine which of the following floating-point constants are VALID for the quantity (5 * 100000).

(a) 500000 (b) 0.5e6 (c) 5E5 (d) 5e5 (e) 5e+5

(f) 500E3 (g) .5E6 (h) 50e4 (i) 50.E+4 (j) 5.0E+5

(k) All of the above (l) None of these

Ans. (k)

[Q006]. What will be the output of the following program :

```
void main()
{
printf("%f",123.);
}
```

(a)123 (b)Compile-Time Error (c)123.00 (d)123.000000

Ans. (d)

[Q007]. What will be the output of the following program :

```
void main()
{
printf("%d",sizeof(integer));
}
```

(a)2 (b)Compile-Time Error (c)4 (d)None of these

Ans. (b) since there is no such data type called 'integer'.

[Q008]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",sizeof str);
}
```

(a)14 (b)Compile-Time Error (c)15 (d)None of these

Ans. (a)

[Q009]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",++(sizeof(str)));
}
```

(a)14 (b)Compile-Time Error (c)15 (d)None of these

Ans. (b)

[Q010]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",-sizeof(str));
}
```

(a)14 (b)Compile-Time Error (c)-15 (d)-14

Ans. (d)

[Q011]. What will be the output of the following program :

```
void main()
{
printf("%d",!(100==100)+1);
}
```

(a)100 (b)0 (c)1 (d)2

Ans. (c)

[Q012]. What will be the output of the following program :

```
void main()
{
int x=5,y=6,z=2;
  z/=y/z==3?y/z:x*y;
  printf("%d",z);
}
```

(a)Compile-Time Error (b)2 (c)0 (d)1

Ans. (c)

[Q013]. What will be the output of the following program :

```
void main()
{
printf("%d %d %d",5,!5,25-!25);
}
```

(a)5 10 22 (b)5 5 25 (c)5 0 25 (d)5 1 24

Ans. (c)

[Q014]. What will be the output of the following program :

```
int main()
```

```
{
  int a=500,b=100,c=30,d=40,e=19;
  a+=b-=c*=d/=e%=5;
  printf("%d %d %d %d %d",a,b,c,d,e);
}
```

(a)500 100 30 40 4 (b)Run-Time Error (c)700 200 300 10 4 (d)300 -200 300 10 4

Ans. (d)

[Q015]. What will be the output of the following program :

```
void main()
{
int a=500,b=100,c=30,d=40,e=19;
  if ((((a > b) ? c : d) >= e) && !((e <= d) ? ((a / 5) == b) : (c == d)))
        printf("Success");
  else
        printf("Failure");
}
```

(a)VALID : Success (b)VALID : Failure (c)INVALID (d)None of these

Ans. (b)

[Q016]. What will be the output of the following program :

```
void main()
{
int a=1,b=2,c=3,d=4;
  printf("%d",!a?b?!c:!d:a);
}
```

(a)1 (b)2 (c)3 (d)4

Ans. (a)

[Q017]. What will be the output of the following program :

```
void main()
{
  int i=12345,j=-13579,k=-24680;
  long ix=123456789;
  short sx=-2222;
  unsigned ux=5555;
  printf("\n%d %d %d %ld %d %u",i,j,k,ix,sx,ux);
  printf("\n\n%3d %3d %3d\n%3ld %3d %3u",i,j,k,ix,sx,ux);
  printf("\n\n%8d %8d %8d\n%15ld %8d %8u",i,j,k,ix,sx,ux);
  printf("\n\n%-8d %-8d\n%-8d %-15ld\n%-8d %-8u",i,j,k,ix,sx,ux);
  printf("\n\n%+8d %+8d\n%+8d %+15ld\n%+8d %8u",i,j,k,ix,sx,ux);
  printf("\n\n%08d %08d\n%08d %015ld\n%08d %08u",i,j,k,ix,sx,ux);
}
```

Ans. 12345 -13579 -24680 123456789 -2222 5555

```
12345 -13579 -24680
123456789 -2222 5555

  12345   -13579   -24680
    123456789    -2222     5555

12345    -13579
-24680   123456789
-2222    5555

 +12345   -13579
 -24680     +123456789
  -2222     5555
```

00012345 -0013579

-0024680 000000123456789

-0002222 00005555

[Q018]. What will be the output of the following program :

```
void main()
{
int i=12345,j=0xabcd9,k=077777;
  printf("%d %x %o",i,j,k);
  printf("\n%3d %3x %3o",i,j,k);
  printf("\n%8d %8x %8o"i,j,k);
  printf("\n%-8d %-8x %-8o",i,j,k);
  printf("\n%+8d %+8x %+8o",i,j,k);
  printf("\n%08d %#8x %#8o",i,j,k);
}
```

Ans. 12345 abcd9 77777

12345 abcd9 77777

12345 abcd9 77777

12345 abcd9 77777

+12345 abcd9 77777

00012345 0xabcd9 077777

[Q019]. What will be the output of the following program :

```
void main()
{
  char c1='A', c2='B', c3='C';
  printf("%c %c %c",c1,c2,c3);
  printf("\n%c%c%c",c1,c2,c3);
  printf("\n%3c %3c %3c",c1,c2,c3);
```

```
        printf("\n%3c%3c%3c",c1,c2,c3);
        printf("\nc1=%c c2=%c c3=%c",c1,c2,c3);
      }
```

Ans. A B C

ABC

A B C

A B C

c1=A c2=B c3=C

[Q020]. What will be the output of the following program :

```
      void main()
      {
        float a=2.5, b=0.0005, c=3000.;
        printf("%f %f %f",a,b,c);
        printf("\n%3f %3f %3f",a,b,c);
        printf("\n%8f %8f %8f",a,b,c);
        printf("\n%8.4f %8.4f %8.4f",a,b,c);
        printf("\n%8.3f %8.3f %8.3f",a,b,c);
        printf("\n%e %e %e",a,b,c);
        printf("\n%3e %3e %3e",a,b,c);
        printf("\n%12e %12e %12e",a,b,c);
        printf("\n%8.2e %8.2e %8.2e",a,b,c);
        printf("\n%-8f %-8f %-8f",a,b,c);
        printf("\n%+8f %+8f %+8f",a,b,c);
        printf("\n%08f %08f %08f",a,b,c);
        printf("\n%#8f %#8f %#8f",a,b,c);
        printf("\n%g %g %g",a,b,c);
        printf("\n%#g %#g %#g"a,b,c);
      }
```

Ans. 2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.5000 0.0005 3000.0000

2.500 0.001 3000.000

2.500000e+000 5.000000e-004 3.000000e+003

2.500000e+000 5.000000e-004 3.000000e+003

2.500000e+000 5.000000e-004 3.000000e+003

2.5000e+000 5.0000e-004 3.0000e+003

2.50e+000 5.00e-004 3.00e+003

2.500000 0.000500 3000.000000

+2.500000 +0.000500 +3000.000000

2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.5 0.0005 3000

2.500000 0.000500 3000.000000

[Q021]. What will be the output of the following program :

```
void main()
{
  char str[]="C For Swimmers";
  printf("%s",str);
  printf("\n%.5s",str);
  printf("\n%8.*s",5,str);
  printf("\n%-10s %.1s",str+6,str);
}
```

Ans. C For Swimmers

C For

C For

Swimmers C

[Q022]. What will be the output of the following program :

```
void main()
{
  int a=1,b=2,c=3;
  scanf("%d %*d %d",&a,&b,&c);
  printf("a=%d b=%d c=%d",a,b,c);
}
```

[NOTE : 3 values entered by the user are:100 200 300]

(a)1 2 3 (b)100 200 300 (c)100 200 3 (d)100 300 3

Ans. (d)

[Q023]. What will be the output of the following program :

```
void main()
{
  char line[80];  // Max. length=80 Chars
  scanf("%[^,]s",line);
  printf("\n%s",line);
}
```

[NOTE : THE USER INPUT IS:Dear Friends, What is the output?]

(a)Compile-Time Error (b)Dear Friends (c)What is the output? (d)None of these

Ans. (b)

[Q024]. What will be the output of the following program :

```
void main()
{
  char a,b,c;
  scanf("%c%c%c",&a,&b,&c);
```

```
  printf("a=%c b=%c c=%c",a,b,c);
 }
```

[NOTE : THE USER INPUT IS :A B C]

(a)a=A b=B c=C (b)a=A b= c=B (c)a=A b= c=C (d)None of these

Ans. (b)

[Q025]. What will be the output of the following program :

```
 void main()
 {
  int i=1;
  float f=2.25;
  scanf("%d a %f",&i,&f);
  printf("%d %.2f",i,f);
 }
```

[NOTE : THE USER INPUT IS:5 5.75]

(a)1 2.25 (b)5 5.75 (c)5 2.25 (d)None of these

Ans. (c)

[Q026]. What will be the output of the following program :

```
 void main()
 {
  char a,b,c;
  scanf("%c %c %c",&a,&b,&c);
  printf("a=%c b=%c c=%c",a,b,c);
 }
```

[NOTE : THE USER INPUT IS :ABC DEF GHI]

(a)a=ABC b=DEF c=GHI (b)a=A b=B c=C (c)a=A b=D c=G (d)None of these

Ans. (b)

[Q027]. What will be the output of the following program :

```
void main()
{
  char a[80],b[80],c[80];
  scanf("%1s %5s %3s",a,b,c);
  printf("%s %s %s",a,b,c);
}
```

[NOTE : THE USER INPUT IS:CMeansSea Ocean Vast]

(a)C O V (b)C Means Sea (c)C Ocean Vas (d)None of these

Ans. (b)

[Q028]. What will be the output of the following program :

```
void main()
{
  int a,b,c;
  scanf("%1d %2d %3d",&a,&b,&c);
  printf("Sum=%d",a+b+c);
}
```

[NOTE : THE USER INPUT IS :123456 44 544]

(a)Sum=480 (b)Sum=594 (c)Sum=589 (d)None of these

Ans. (a)

[Q029]. What happens when the following program is executed :

```
void main()
{
  char line[80];
```

```
  scanf("%[^1234567890\n]",line);
}
```

(a)Accepts the string that contains DIGITS only.

(b)Accepts the string that contains DIGITS and NEWLINE characters.

(c)Accepts the string that contains anything other than the DIGITS and NEWLINE characters.

(d)None of these

Ans. (c)

[Q030]. What happens when the following program is executed :

```
void main()
{
  char line[80];
scanf("%[^*]",line);
}
```

(a)Accepts the string that contains DIGITS & ALPHABETS only.

(b)Accepts the string that contains * or asterisk characters only.

(c)Accepts the string that contains anything other than the * or asterisk character.

(d)None of these

Ans. (c)

[Q001]. Determine which of the following are VALID identifiers. If invalid, state the reason.

(a) sample1 (b) 5sample (c) data_7 (d) return (e) #fine

(f) variable (g) 91-080-100 (h) name & age (i) _val (j) name_and_age

Ans. (a) VALID

(b) Invalid, since an identifier must begin with a letter or an underscore

(c) VALID

(d) Invalid, since return is a reserved word

(e) Invalid, since an identifier must begin with a letter or an underscore

(f) VALID

(g) Invalid, since an identifier must begin with a letter or an underscore

(h) Invalid, since blank spaces are not allowed

(i) VALID

(j) VALID

[Q002]. Determine which of the following are VALID character constants. If invalid, state the reason.

(a) 'y' (b) '\r' (c) 'Y' (d) '@' (e) '/r'

(f) 'word' (g) '\0' (h) '\?' (i) '\065' (j) '\'' (k) ' '

Ans. (a) VALID

(b) VALID

(c) VALID

(d) VALID

(e) Invalid, since escape sequences must be written with a backward slash (i.e. \)

(f) Invalid, since a character constant cannot consist of multiple characters

(g) VALID (null-character escape sequence)

(h) VALID

(i) VALID (Octal escape sequence)

(j) VALID

(k) VALID

[Q003]. Determine which of the following are VALID string constants. If invalid, state the reason.

(a) 'Hi Friends' (b) "abc,def,ghi" (c) "Qualification

(d) "4325.76e-8" (e) "Don\'t sleep" (f) "He said, "You\'re great"

(g) "" (h) " " (i) "Rs.100/-"

Ans. (a) Invalid, since a string constant must be enclosed in double quotation marks

(b) VALID

(c) Invalid, since trailing quotation mark is missing

(d) VALID

(e) VALID (single-quote escape sequence)

(f) Invalid, since the quotation marks and (optionally) apostrophe within the string cannot be expressed without the escape sequences.

(g) VALID

(h) VALID

(i) VALID

[Q004]. Determine which of the following numerical values are valid constants. If a constant is

valid, specify whether it is integer or real. Also, specify the base for each valid integer constant.

(a) 10,500 (b) 080 (c) 0.007 (d) 5.6e7 (e) 5.6e-7

(f) 0.2e-0.3 (g) 0.2e 0.3 (h) 0xaf9s82 (i) 0XABCDEFL (j) 0369CF

(k) 87654321l (l) 87654321

Ans. (a) Invalid, since illegal character(,)

(b) VALID

(c) VALID

(d) VALID

(e) VALID

(f) VALID

(g) Invalid, since illegal character(blank space)

(h) Invalid, since illegal character(s)

(i) VALID

(j) Invalid, since illegal characters (9, C, F), if intended as an octal constant.

(k) VALID

(l) VALID

[Q005]. Determine which of the following floating-point constants are VALID for the quantity (5 * 100000).

(a) 500000 (b) 0.5e6 (c) 5E5 (d) 5e5 (e) 5e+5

(f) 500E3 (g) .5E6 (h) 50e4 (i) 50.E+4 (j) 5.0E+5

(k) All of the above (l) None of these

Ans. (k)

[Q006]. What will be the output of the following program :

```
void main()
{
printf("%f",123.);
}
```

(a)123 (b)Compile-Time Error (c)123.00 (d)123.000000

Ans. (d)

[Q007]. What will be the output of the following program :

```
void main()
{
printf("%d",sizeof(integer));
}
```

(a)2 (b)Compile-Time Error (c)4 (d)None of these

Ans. (b) since there is no such data type called 'integer'.

[Q008]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",sizeof str);
```

```
}
```

(a)14 (b)Compile-Time Error (c)15 (d)None of these

Ans. (a)

[Q009]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",++(sizeof(str)));
}
```

(a)14 (b)Compile-Time Error (c)15 (d)None of these

Ans. (b)

[Q010]. What will be the output of the following program :

```
void main()
{
char str[]="C For Swimmers";
  printf("%d",-sizeof(str));
}
```

(a)14 (b)Compile-Time Error (c)-15 (d)-14

Ans. (d)

[Q011]. What will be the output of the following program :

```
void main()
{
printf("%d",!(100==100)+1);
```

```
}
```

(a)100 (b)0 (c)1 (d)2

Ans. (c)

[Q012]. What will be the output of the following program :

```
void main()
{
int x=5,y=6,z=2;
  z/=y/z==3?y/z:x*y;
  printf("%d",z);
}
```

(a)Compile-Time Error (b)2 (c)0 (d)1

Ans. (c)

[Q013]. What will be the output of the following program :

```
void main()
{
printf("%d %d %d",5,!5,25-!25);
}
```

(a)5 10 22 (b)5 5 25 (c)5 0 25 (d)5 1 24

Ans. (c)

[Q014]. What will be the output of the following program :

```
int main()
{
  int a=500,b=100,c=30,d=40,e=19;
  a+=b-=c*=d/=e%=5;
```

```
    printf("%d %d %d %d %d",a,b,c,d,e);
  }
```

(a)500 100 30 40 4 (b)Run-Time Error (c)700 200 300 10 4 (d)300 -200 300 10 4

Ans. (d)

[Q015]. What will be the output of the following program :

```
  void main()
  {
 int a=500,b=100,c=30,d=40,e=19;
    if ((((a > b) ? c : d) >= e) && !((e <= d) ? ((a / 5) == b) : (c == d)))
        printf("Success");
    else
        printf("Failure");
  }
```

(a)VALID : Success (b)VALID : Failure (c)INVALID (d)None of these

Ans. (b)

[Q016]. What will be the output of the following program :

```
  void main()
  {
 int a=1,b=2,c=3,d=4;
    printf("%d",!a?b?!c:!d:a);
  }
```

(a)1 (b)2 (c)3 (d)4

Ans. (a)

[Q017]. What will be the output of the following program :

```
void main()
{
  int i=12345,j=-13579,k=-24680;
  long ix=123456789;
  short sx=-2222;
  unsigned ux=5555;
  printf("\n%d %d %d %ld %d %u",i,j,k,ix,sx,ux);
  printf("\n\n%3d %3d %3d\n%3ld %3d %3u",i,j,k,ix,sx,ux);
  printf("\n\n%8d %8d %8d\n%15ld %8d %8u",i,j,k,ix,sx,ux);
  printf("\n\n%-8d %-8d\n%-8d %-15ld\n%-8d %-8u",i,j,k,ix,sx,ux);
  printf("\n\n%+8d %+8d\n%+8d %+15ld\n%+8d %8u",i,j,k,ix,sx,ux);
  printf("\n\n%08d %08d\n%08d %015ld\n%08d %08u",i,j,k,ix,sx,ux);
}
```

Ans. 12345 -13579 -24680 123456789 -2222 5555

```
12345 -13579 -24680
123456789 -2222 5555

  12345   -13579   -24680
    123456789    -2222     5555

12345    -13579
-24680   123456789
-2222    5555

 +12345   -13579
-24680      +123456789
 -2222     5555

00012345 -0013579
-0024680 000000123456789
-0002222 00005555
```

[Q018]. What will be the output of the following program :

```
void main()
{
int i=12345,j=0xabcd9,k=077777;
  printf("%d %x %o",i,j,k);
  printf("\n%3d %3x %3o",i,j,k);
  printf("\n%8d %8x %8o"i,j,k);
  printf("\n%-8d %-8x %-8o",i,j,k);
  printf("\n%+8d %+8x %+8o",i,j,k);
  printf("\n%08d %#8x %#8o",i,j,k);
}
```

Ans. 12345 abcd9 77777

```
12345 abcd9 77777
  12345    abcd9    77777
12345    abcd9    77777
 +12345    abcd9    77777
00012345  0xabcd9   077777
```

[Q019]. What will be the output of the following program :

```
void main()
{
  char c1='A', c2='B', c3='C';
  printf("%c %c %c",c1,c2,c3);
  printf("\n%c%c%c",c1,c2,c3);
  printf("\n%3c %3c %3c",c1,c2,c3);
  printf("\n%3c%3c%3c",c1,c2,c3);
  printf("\nc1=%c c2=%c c3=%c",c1,c2,c3);
}
```

Ans. A B C

ABC

A B C

A B C

c1=A c2=B c3=C

[Q020]. What will be the output of the following program :

```
void main()
{
  float a=2.5, b=0.0005, c=3000.;
  printf("%f %f %f",a,b,c);
  printf("\n%3f %3f %3f",a,b,c);
  printf("\n%8f %8f %8f",a,b,c);
  printf("\n%8.4f %8.4f %8.4f",a,b,c);
  printf("\n%8.3f %8.3f %8.3f",a,b,c);
  printf("\n%e %e %e",a,b,c);
  printf("\n%3e %3e %3e",a,b,c);
  printf("\n%12e %12e %12e",a,b,c);
  printf("\n%8.2e %8.2e %8.2e",a,b,c);
  printf("\n%-8f %-8f %-8f",a,b,c);
  printf("\n%+8f %+8f %+8f",a,b,c);
  printf("\n%08f %08f %08f",a,b,c);
  printf("\n%#8f %#8f %#8f",a,b,c);
  printf("\n%g %g %g",a,b,c);
  printf("\n%#g %#g %#g"a,b,c);
}
```

Ans. 2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.5000 0.0005 3000.0000

2.500 0.001 3000.000

2.500000e+000 5.000000e-004 3.000000e+003

2.500000e+000 5.000000e-004 3.000000e+003

2.500000e+000 5.000000e-004 3.000000e+003

2.5000e+000 5.0000e-004 3.0000e+003

2.50e+000 5.00e-004 3.00e+003

2.500000 0.000500 3000.000000

+2.500000 +0.000500 +3000.000000

2.500000 0.000500 3000.000000

2.500000 0.000500 3000.000000

2.5 0.0005 3000

2.500000 0.000500 3000.000000

[Q021]. What will be the output of the following program :

```
void main()
{
  char str[]="C For Swimmers";
  printf("%s",str);
  printf("\n%.5s",str);
  printf("\n%8.*s",5,str);
  printf("\n%-10s %.1s",str+6,str);
}
```

Ans. C For Swimmers

C For

C For

Swimmers C

[Q022]. What will be the output of the following program :

```
void main()
```

```
{
  int a=1,b=2,c=3;
  scanf("%d %*d %d",&a,&b,&c);
  printf("a=%d b=%d c=%d",a,b,c);
}
```

[NOTE : 3 values entered by the user are:100 200 300]

(a)1 2 3 (b)100 200 300 (c)100 200 3 (d)100 300 3

Ans. (d)

[Q023]. What will be the output of the following program :

```
void main()
{
  char line[80];  // Max. length=80 Chars
  scanf("%[^,]s",line);
  printf("\n%s",line);
}
```

[NOTE : THE USER INPUT IS:Dear Friends, What is the output?]

(a)Compile-Time Error (b)Dear Friends (c)What is the output? (d)None of these

Ans. (b)

[Q024]. What will be the output of the following program :

```
void main()
{
  char a,b,c;
  scanf("%c%c%c",&a,&b,&c);
  printf("a=%c b=%c c=%c",a,b,c);
}
```

[NOTE : THE USER INPUT IS :A B C]

(a)a=A b=B c=C (b)a=A b= c=B (c)a=A b= c=C
(d)None of these

Ans. (b)

[Q025]. What will be the output of the following program :

```
void main()
{
  int i=1;
  float f=2.25;
  scanf("%d a %f",&i,&f);
  printf("%d %.2f",i,f);
}
```

[NOTE : THE USER INPUT IS:5 5.75]

(a)1 2.25 (b)5 5.75 (c)5 2.25 (d)None of these

Ans. (c)

[Q026]. What will be the output of the following program :

```
void main()
{
  char a,b,c;
  scanf("%c %c %c",&a,&b,&c);
  printf("a=%c b=%c c=%c",a,b,c);
}
```

[NOTE : THE USER INPUT IS :ABC DEF GHI]

(a)a=ABC b=DEF c=GHI (b)a=A b=B c=C (c)a=A b=D c=G
(d)None of these

Ans. (b)

[Q027]. What will be the output of the following program :

```
void main()
{
  char a[80],b[80],c[80];
  scanf("%1s %5s %3s",a,b,c);
  printf("%s %s %s",a,b,c);
}
```

[NOTE : THE USER INPUT IS:CMeansSea Ocean Vast]

(a)C O V (b)C Means Sea (c)C Ocean Vas (d)None of these

Ans. (b)

[Q028]. What will be the output of the following program :

```
void main()
{
  int a,b,c;
  scanf("%1d %2d %3d",&a,&b,&c);
  printf("Sum=%d",a+b+c);
}
```

[NOTE : THE USER INPUT IS :123456 44 544]

(a)Sum=480 (b)Sum=594 (c)Sum=589 (d)None of these

Ans. (a)

[Q029]. What happens when the following program is executed :

```
void main()
{
  char line[80];
  scanf("%[^1234567890\n]",line);
}
```

(a)Accepts the string that contains DIGITS only.

(b)Accepts the string that contains DIGITS and NEWLINE characters.

(c)Accepts the string that contains anything other than the DIGITS and NEWLINE characters.

(d)None of these

Ans. (c)

[Q030]. What happens when the following program is executed :

```
void main()
{
  char line[80];
scanf("%[^*]",line);
}
```

(a)Accepts the string that contains DIGITS & ALPHABETS only.

(b)Accepts the string that contains * or asterisk characters only.

(c)Accepts the string that contains anything other than the * or asterisk character.

(d)None of these

Ans. (c)

. How many destructors can a class have?

(A) 0 (B) 1 (C) 2 (D) None of the above

2. Which of the following statements is TRUE?

(A) The iostream class is a subclass of the fstream class.

(B) The iostream class is a subclass of both the istream class and the ostream class.

(C) The iostream class is a subclass of the ifstream class.

(D) None of the above.

3. Which members of a base class are inherited by a subclass?

(A) All the public and private members. (B) All the protected and private members.

(C) All the public and protected members. (D) None of the above.

4. In C++, a function contained within a class is called

(A) a member function. (B) an operation. (C) a class function. (D) a method.

5. Overloading a postfix increment operator by means of a member function takes

(A) no argument. (B) one argument. (C) two argument. (D) three argument.

6. The main( ) function returns value of type

(A) real. (B) character. (C) int. (D) null.

7. The library function exit () causes an exit from

(A) the loop in which it occurs. (B) the block in which it occurs.
(C) the function in which it occurs. (D) the program in which it occurs.

8. What would be the output for the following program?

```
int main( )
{
  int x, y=10,z=10;
  x=(y==z);
 cout<<x;
 return 0;
}
```

(A) 0 (B) 1 (C) 10 (D) Error

9. An exception is caused by

(A) a hardware problem. (B) a problem in the operating system.
(C) a syntax error. (D) a run time error.

10. A white space is:

(A) blank space. (B) new line. (C) tab. (D) all of the above.

11. The sizeof() operator accepts _____ parameter.

(A) no (B) more than one (C) size (D) one

12. C++ supports_____.

(A) only call-by-value (B) only call-by-reference
(C) both call-by-value and call-by-reference (D) call-by-function

13. The typeid operator allows the type of an expression to be determined at _____.

(A) run-time (B) compile-time (C) random (D) error-time

14. The loop for (; ;) will

(A) Executes only one time (B) Create infinite loop (C) Not executes (D) Syntax error

15. The fields in a class of a C++ program are by default _____.
(A) Protected (B) Public (C) Private (D) None of these

16. _____are bound dynamically at run time.
(A) static class (B) virtual functions (C) friend functions (D) inline functions

17. Which of these C++ feature allow you to create classes that are dynamic in terms of the types of data they can handle?
(A) Inheritance (B) Templates (C) Polymorphism (D) Information hiding

18.
```
int a = 10;
void main()
{
   int a = 20;
  cout << a << ::a;
}
```
The output of this program is
(A) 10 20 (B) 20 10 (C) Syntax error (D) 20 20

19. Usually a pure virtual function
(A) has complete function body. (B) will never be called. (C) will be called only to delete an object.
(D) is defined only in derived class.

20. Overloading the function operator
(A) requires a class with an overloaded operator.
(B) requires a class with an overloaded [ ] operator.
(C) allows you to create objects that act syntactically like functions.
(D) usually make use of a constructor that takes arguments.

21. If an array is declared as int a[4] = {3, 0, 1, 2}, then values assigned to a[0] & a[4] will be _____
(A) 3, 2 (B) 0, 2 (C) 3, 0 (D) 0, 4

22. In C++, dynamic memory allocation is accomplished with the operator ____
(A) new (B) this (C) malloc( ) (D) delete

23. The keyword *friend* does not appear in
(A) the class allowing access to another class. (B) the class desiring access to another class.
(C) the private section of a class. (D) the public section of a class.

24. A friend function to a class, C cannot access
(A) private data members and member functions.
(B) public data members and member functions.
(C) protected data members and member functions.
(D) the data members of the derived class of C.

25. The operator that cannot be overloaded is

(A) ++ (B) :: (C) ( ) (D) ~

26. *this* pointer

(A) implicitly points to an object. (B) can be explicitly used in a class.
(C) can be used to return an object. (D) All of the above.

27. Data members which are *static*

(A) cannot be assigned a value (B) can only be used in *static* functions
(C) cannot be defined in a *Union* (D) can be accessed outside the class

28. It is possible to declare as a *friend*

(A) a member function (B) a global function (C) a class (D) all of the above

29. In multiple inheritance

(A) the base classes must have only default constructors (B) cannot have virtual functions
(C) can include virtual classes (D) None of the above.

30. A *virtual* class is the same as

(A) an abstract class (B) a class with a virtual function (C) a base class (D) none of the above.

31. Consider the following statements

char *ptr;
ptr = "hello";
cout << *ptr;
What will be printed?

(A) first letter (B) entire string (C) it is a syntax error (D) last letter

32. In which case is it mandatory to provide a destructor in a class?

(A) Almost in every class (B) Class for which two or more than two objects will be created
(C) Class for which copy constructor is defined
(D) Class whose objects will be created dynamically

33. Exception handling is targeted at

(A) Run-time error (B) Compile time error (C) Logical error (D) All of the above.

34. A pointer to the base class can hold address of

(A) only base class object (B) only derived class object
(C) base class object as well as derived class object (D) None of the above

35. Function templates can accept

(A) any type of parameters (B) only one parameter (C) only parameters of the basic type
(D) only parameters of the derived type

36. Which of the following is the valid class declaration header for the derived class d with base classes b1 and b2?

(A) class d : public b1, public b2 (B) class d : class b1, class b2 (C) class d : public b1, b2
(D) class d : b1, b2

37. A library function exit() causes an exit from

(A) the loop in which it occurs (B) the block in which it occurs
(C) the function in which it occurs (D) the program in which it occurs

38. A pure virtual function is a virtual function that

(A) has no body (B) returns nothing (C) is used in base class (D) both (A) and (C)

39. A static function

(A) should be called when an object is destroyed.
(B) is closely connected with and individual object of a class.
(C) can be called using the class name and function name.
(D) is used when a dummy object must be created.

40. Which of the following is not the characteristic of constructor.

(A) They should be declared in the public section. (B) They do not have return type.
(C) They can not be inherited. (D) They can be virtual.

41. Name the header file to be included for the use of built in function isalnum()

(A) string.h (B) process.h (C) ctype.h (D) dos.h

42. What is the output of given code fragment?

```
int f=1, i=2;
while(++i<5)
f*=i;
cout<<f;
```

(A) 12 (B) 24 (C) 6 (D) 3

43. What will be the values of x, m and n after the execution of the following statements?

```
int x, m, n;
m = 10;
n = 15;
x = ++m + n++;
```

(A) x=25, m=10, n=15 (B) x=26, m=11, n=16 (C) x=27, m=11, n=16 (D) x=27, m=10, n=15

44. Which of the following will produce a value 10 if x = 9.7?

(A) floor(x) (B) abs(x) (C) log(x) (D) ceil(x)

45. Which of the following statements are true in c++?

(A) Classes can not have data as public members. (B) Structures cannot have functions as members.

(C) Class members are public by default. (D)None of these.

46. What is the error in the following code?

```
class t
{
virtual void print();
}
```

(A) No error. (B) Function print() should be declared as static.

(C) Function print() should be defined. (D) Class t should contain data members.

47. What would be the output of the following?

```
#include<iostream.h>
void main()
{
char *ptr="abcd"
char ch;
ch = ++*ptr++;
cout<<ch;
}
```

(A) a (B) b (C) c (D) d

48. A copy constructor takes

(A) no argument (B) one argument (C) two arguments (D) arbitrary no. of arguments

49. Which of the following ways are legal to access a class data member using this pointer?

(A) this.x (B) *this.x (C) *(this.x) (D) (*this).x

50. Which of the following operator can be overloaded through friend function?

(A) -> (B) = (C) ( ) (D) *

51. Maximum number of template arguments in a function template is

(A) one (B) two (C) three (D) many

52. In access control in a protected derivation, visibility modes will change as follows:

(A) private, public and protected become protected (B) only public becomes protected.

(C) public and protected become protected. (D) only private becomes protected.

1. For a binary tree, the maximum number of nodes at level i will be _____.

(A) $2^{i+1} - 1$ (B) 2^{i} (C) 2^{i-1} (D) none

2. The order of the bubble sort algorithm is _____.

(A) $O(n^2)$ (B) O(n) (C) first (D) second

3. A leaf node of a tree is a node with a degree equal to _____.

(A) 0 (B) 1 (C) 2 (D) k

4. An AVL tree of height *h* has utmost the following number of vertices

(A) $2^{h-1} - 1$ (B) $2^{h-1} + 1$ (C) $2^{h+1} + 1$ (D) $2^{h+1} - 1$

5. Extra memory of *O(n)* is needed in

(A) Insertion Sort. (B) Bubble Sort. (C) Merge Sort. (D) Quick Sort.

6. The worst case complexity of binary search tree is

(A) $O(2n)$. (B) $O(n/2)$. (C) $O(n^2)$. (D) $O(n)$.

7. Applications of Stacks are

(A) Library Management Systems. (B) Job Scheduling in operating systems

(C) Compilers and postfix and prefix evaluations. (D) Multiple precision arithmetic.

8. In Dijkstra's algorithm

(A) Negative weight edges are sometimes used. (B) Negative weight edges are not used.

(C) Negative weight edges are set to zero. (D) Negative weight edges are always used.

9. Worst case performance of Quick sort can happen if

(A) Input array is unsorted. (B) Input array is partially sorted.

(C) Input array is empty. (D) Input array is sorted.

10. In the following hash function method, key is divided into several parts, further combined

together and transformed in a certain way to create the target address.

(A) Folding (B) Mid Square (C) Division (D) Extraction

11. A technique for direct search is

(A) Binary Search (B) Linear Search (C) Tree Search (D) Hashing

12. The searching technique that takes O (1) time to find a data is

(A) Linear Search (B) Binary Search (C) Hashing (D) Tree Search

13. The number of interchanges required to sort 5, 1, 6, 2 4 in ascending order using Bubble Sort

is

(A) 6 (B) 5 (C) 7 (D) 8

14. The postfix form of the expression (A+ B)*(C*D– E)*F / G is

(A) AB+ CD*E – FG /** (B) AB + CD* E – F **G / (C) AB + CD* E – *F *G /
(D) AB + CDE * – * F *G /

15. In worst case Quick Sort has order

(A) O (n log n) (B) O (n2/2) (C) O (log n) (D) O (n2/4)

16. A full binary tree with 2n+1 nodes contain

(A) n leaf nodes (B) n non-leaf nodes (C) n-1 leaf nodes (D) n-1 non-leaf nodes

17. Which of the following sorting algorithms does not have a worst case running time of (2) O n ?

(A) Insertion sort (B) Merge sort (C) Quick sort (D) Bubble sort

18. Consider a linked list of n elements. What is the time taken to insert an element after an

element pointed by some pointer?

(A) O (1) (B) O (log2 n) (C) O (n) (D) O (n log2 n)

19. The maximum degree of any vertex in a simple graph with *n* vertices is

(A) *n–1* (B) *n+1* (C) *2n–1* (D) *n*

20. The data structure required for Breadth First Traversal on a graph is

(A) queue (B) stack (C) array (D) tree

21. The quick sort algorithm exploit _____ design technique

(A) Greedy (B) Dynamic programming (C) Divide and Conquer (D) Backtracking

22. The number of different directed trees with 3 nodes are

(A) 2 (B) 3 (C) 4 (D) 5

23. One can convert a binary tree into its mirror image by traversing it in

(A) inorder (B) preorder (C) postorder (D) any order

24. The number of leaf nodes in a complete binary tree of depth d is

(A) 2d (B) 2d–1+1 (C) 2d+1+1 (D) 2d+1

25. The complexity of searching an element from a set of n elements using Binary search algorithm is

(A) O(n) (B) O(log n) (C) O(n2) (D) O(n log n)

26. What data structure would you mostly likely see in a nonrecursive implementation of a recursive algorithm?

(A) Stack (B) Linked list (C) Queue (D) Trees

27. A BST is traversed in the following order recursively: Right, root, left

The output sequence will be in

(A) Ascending order (B) Descending order (C) Bitomic sequence (D) No specific order

28. An adjacency matrix representation of a graph cannot contain information of :

(A) nodes (B) edges (C) direction of edges (D) parallel edges

29. One of the major drawback of B-Tree is the difficulty of traversing the keys sequentially.

(A) True (B) False

30. What is the maximum number of nodes in a heap with 8 leaf nodes?

(A) 15 (B) 16 (C) 17 (D) 31

31. A binary tree with 27 nodes has _____ null branches.

(A) 54 (B) 27 (C)26 (D) None of the above

32. Which of the following data structure is linear type?

A. Strings B. Queue C. Lists D. All of the above

33. Number of binary trees formed with 5 nodes are

A. 30 B. 36 C. 108 D. 42

34. What is Dequeue?

A. Elements can be added from front

B. Elements can be added to or removed from either the front or rear

C. Elements can be added from rear D. None of the above

35. Which of the following search algorithm requires a sorted array?

A. Linear search B. Hash search C. Binary search D. All of these

36 . A directed graph is ……. if there is a path from each vertex to every other vertex in the

digraph.

A) Weakly connected B) Strongly Connected C) Tightly Connected D) Linearly Connected

37. In the …….. traversal we process all of a vertex's descendents before we move to an

adjacent vertex.

A) Depth First B) Breadth First C) With First D) Depth Limited

38. State True of False.

i) Network is a graph that has weights or costs associated with it.

ii) An undirected graph which contains no cycles is called a forest.

iii) A graph is said to be complete if there is no edge between every pair of vertices.

A) True, False, True B) True, True, False C) True, True, True D) False, True, True

39. Match the following.

a) Completeness i) How long does it take to find a solution

b) Time Complexity ii) How much memory need to perform the search.

c) Space Complexity iii) Is the strategy guaranteed to find the solution when there in one.

A) a-iii, b-ii, c-I B) a-i, b-ii, c-iii C) a-iii, b-i, c-ii D) a-i, b-iii, c-ii

40. The number of comparisons done by sequential search is ……

A) (N/2)+1 B) (N+1)/2 C) (N-1)/2 D) (N+2)/2

41. In ……, search start at the beginning of the list and check every element in the list.

A) Linear search B) Binary search C) Hash Search D) Binary Tree search

42. State True or False.

i) Binary search is used for searching in a sorted array.

ii) The time complexity of binary search is O(logn).

A) True, False B) False, True C) False, False D) True, True

43. Which of the following is not the internal sort?

A) Insertion Sort B) Bubble Sort C) Merge Sort D) Heap Sort

44. State True or False.

i) An undirected graph which contains no cycles is called forest.

ii) A graph is said to be complete if there is an edge between every pair of vertices.

A) True, True B) False, True C) False, False D) True, False

45. A graph is said to be …… if the vertices can be split into two sets V1 and V2 such there are no edges between two vertices of V1 or two vertices of V2.

A) Partite B) Bipartite C) Rooted D) Bisects

46. In a queue, the initial values of front pointer f rare pointer r should be …….. and ……. respectively.

A) 0 and 1 B) 0 and -1 C) -1 and 0 D) 1 and 0

47. In a circular queue the value of r will be ..

A) r=r+1 B) r=(r+1)% [QUEUE_SIZE – 1] C) r=(r+1)% QUEUE_SIZE
D) r=(r-1)% QUEUE_SIZE

48. Which of the following statement is true?

i) Using singly linked lists and circular list, it is not possible to traverse the list backwards.

ii) To find the predecessor, it is required to traverse the list from the first node in case of singly linked list.

A) i-only B) ii-only C) Both i and ii D) None of both

49. The advantage of ……. is that they solve the problem if sequential storage representation. But disadvantage in that is they are sequential lists.

A) Lists B) Linked Lists C) Trees D) Queues

50. What will be the value of top, if there is a size of stack STACK_SIZE is 5

A) 5 B) 6 C) 4 D) None

51. …… is not the operation that can be performed on queue.

A) Insertion B) Deletion C) Retrieval D) Traversal

52. There is an extra element at the head of the list called a …….

A) Antinel B) Sentinel C) List header D) List head

53. A graph is a collection of nodes, called ……. And line segments called arcs or …….. that connect pair of nodes.

A) vertices, edges B) edges, vertices C) vertices, paths D) graph node, edges

54. A ……. is a graph that has weights of costs associated with its edges.

A) Network B) Weighted graph C) Both A and B D) None A and B

55. In general, the binary search method needs no more than ……. comparisons.

A) $[\log_2 n]-1$ B) $[\log n]+1$ C) $[\log_2 n]$ D) $[\log_2 n]+1$

56. Arrays are best data structures

A) for relatively permanent collections of data
B) for the size of the structure and the data in the structure are constantly changing
C) for both of above situation

57. The disadvantage in using a circular linked list is .....

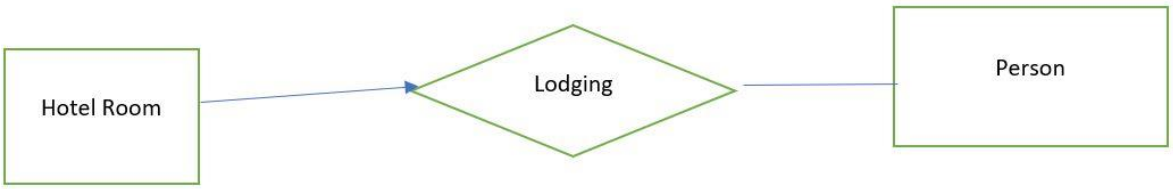
A) It is possible to get into infinite loop. B) Last node points to first node.
C) Time consuming D) Requires more memory space

58. A …… is an acyclic digraph, which has only one node with indegree 0, and other nodes have

indegree 1.

A) Directed tree B) Undirected tree C) Dis-joint tree D) Direction oriented tree

1. If you take these entities 'hotel room', and 'person' with a many-to-many relationship, 'lodging' as shown in the figure below.



To store information about the rent payment to be made by person(s) occupying different hotel rooms, then this information should appear as an attribute of_____

a. Hotel Room

b. Person

c. Lodging

d. None of the Above

2. From the basic ER and relational models given below, which of the statements below is "Incorrect"?

a. An attribute of an entity can be composite

b. In a row of a relational table, an attribute can have exactly one or Null value

c. An attribute of an entity can have more than one value

d. In a row of relational table, an attribute can have more than one value

3. Which of the given statements is considered "TRUE" about an SQL query?

P: An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause

Q: An SQL query can contain a HAVING clause only if it has a GROUP BY clause

R: All attributes used in the GROUP BY clause must appear in the SELECT clause

S: Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

a. P and R

b. P and S

c. Q and R

d. Q and S

4. An instance of a relational scheme R(A, B, C) is given with distinct values for attribute A. Can you conclude that A is a candidate key for R?

(a) True

(b) False

5. Which normal form is considered adequate for normal relational database design?

(a) 3 N F

(b) 5 N F

(c) 2 N F

(d) 4 N F

6**.** The following table has two attributes, A and C, where A is the primary key and C is the foreign key referencing A with on-delete cascade.

A	**B**
2	**4**
3	**4**
4	**3**
5	**2**
7	**2**
9	**5**
6	**4**

The set of all tuples that must be additionally deleted to preserve referential integrity. When the tuple (2,4) is deleted is:

(a) (3, 4) and (6, 4)

(b) (5,2) and (7,2)

(c) (5,2), (7,2) and (9,5)

(d) (3,4), (4,3) and (6,4)

7. Which of these statements below is/ are correct?

(a) SQL permits attribute names to be repeated in the same relation

(b) SQL query automatically eliminates duplicates

(c) SQL query will work if there are no indexes on the relations

(d) None of the above-mentioned

8. Given that the order of an internal node in a B^+ tree, index is the maximum number of children it can have. Imagine a child pointer takes 6 bytes, the search field value takes 14 bytes, and the block size is 512 bytes. Then, what is the order of the internal node?

(a) 27

(b) 26

(c) 25

(d) 24

9. Referential Integrity constraints functions using the _____ concept

(a) Primary key

(b) Super key

(c) Foreign key

(d) Secondary key

10. From the statements mentioned below, which of them may result in an irrevocable error in a database system?

(a) Transaction reads a data item after it is written by an uncommitted transaction

(b) Transaction reads a data item after it is written by a committed transaction

(c) Transaction writes a data item after it is read by a committed transaction

(d) Transaction writes a data item after it is read by an uncommitted transaction

11. In a relational schema, every tuple is divided into fields, known as _____

(a) Queries

(b) Domains

(c) Relations

(d) None of the above

12. _____ is given as the logical design of the database, while the snapshot of the data in the database at a point in time is_____

(a) Database relation, attribute

(b) Database schema, attribute domain

(c) Database schema, database instance

(d) Attribute domain, attribute value

13. The set of attributes X will be fully functionally dependent on the set of attributes Y, if one of the conditions given below is met. What is it?

a. X is not functionally dependent on any sub-set of Y

b. X is functionally dependent on Y

c. None

d. both(a) and (b)

14. _____ is an aggregate function in SQL.

a. Ordered by

b. Distinct

c. Avg

d. Select

15. _____ of the following in place sorting algorithms needs the minimum number of swaps.

a. Selection sort

b. Insertion sort

c. Heap sort

d. Quick sort

16. What does an Embedded pointer provide?

a. A physical record key

b. A prime key

c. An inverted index

d. A secondary access path

17. A hashing function in a Hierarchical database is used to locate the_____

a. Records

b. Collision

c. Foreign key

d. Root

18. Which among the following is a true statement?

(i) Lossless, dependency preserving decomposition into 3NF is always possible

(ii) Any relation with two attributes is in BCNF

a. (i)

b. (ii)

c. (i) and (ii)

d. None of the above

19. Which is the maximum number of super keys for the relation schema R(X,Y,Z,W) with X as the key?

(a)12

(b)6

(c)8

(d)9

20. Which of the following is not part of the ACID properties?

a. Consistency

b. Atomicity

c. Inconsistency

d. Isolation

21. The fields on which clustering index is defined are of type_____

a. Key and non-ordering

b. Non-key and ordering

c. Key and ordering

d. Non-key and non-ordering

22. What is the command used to remove a relation from an SQL database?

a. Remove table

b. Update table

c. Drop table

d. Delete table

23. Every functional dependency that is in set E and also in closure of F is classified as_____

a. F plus is covered by E

b. E is covered by F

c. FD is covered by E

d. F is covered by E

24. What is “AS” clause in SQL used for?

a. Selection Operation

b. Join Operation

c. Projection Operation

d. Rename Operation

25. If a block can hold either 3 records or 10 key pointers and a database contains “n” records, then how many blocks do we need to hold the data file and the dense index?

(a)13n/30

(b) N/10

(c) N/30

(d) N/3

26. From the given desired features, which are beyond the capability of relational algebra?

(a) Multiplication

(b) Finding transitive closure

(c) Aggregate computation

(d) All the above

27. In conservative two phase locking protocol, a transaction_____

a. Should release exclusive locks only post the commit operation

b. Should acquire all the locks at the beginning of transaction

c. Should acquire all the exclusive locks at the beginning of transaction

d. Should release all the locks only at beginning of transaction

28. _____ is not modification of database.

a. Updating

b. Deletion

c. Sorting

d. Insertion

29. _____ is based on Multi Valued Dependency.

a. First

b. Second

c. Third

d. Fourth

30. _____ is the operation to increase the length of a list.

a. Look-up

b. Modify

c. Insert

d. None of these

31. What refers to the completeness and the correctness of the data in a database?

a. Data integrity

b. Data security

c. Data independence

d. Data constraint

32. DML (Data Manipulation Language) is not used for_____

a. Insertion of new information into the Database

b. Modification of information in the Database

c. Deletion of information in the Database

d. Creation of information table in the Database

33. What is a report generator used for?

a. Print files on paper

b. Update files

c. Data entry

d. None of these

34. What are database management systems intended for?

a. Manage file access

b. Establish relationships between records in different files

c. Delete data redundancy

d. None of the above

35. How many types of keys are there in Database Design?

a. Primary key

b. Candidate key

c. Foreign key

d. All the above

36. What does a scheme describe?

a. Records and files

b. Data elements

c. Record relationships

d. All of these

37. What does FD in DBMS stand for?

a. Functional Data

b. Facilitate Data

c. Functional Dependency

d. Facilitate Dependency

38. Row, in Mathematical term is referred as_____

a. Tuple

b. Relation

c. Domain

d. Attribute

39. What do ODBC stand for?

a. Oriented database connection

b. Offline database connection

c. Open database connection

d. None of these

40. Queries to the database is_____

a. Written in English

b. Can aggregate functions such as SUM and COUNT

c. Both (a) and (b)

d. None of these

41. A program is a/an___ entity.

a. Dormant

b. Passive

c. Hyperactive

d. Active

42. Data warehouse supports_____

a. OLTP

b. OLAP

c. Operational Database

d. both (a) and (b)

43. The Classification rules in Data Mining are extracted from_____

a. Information
b. Database
c. Decision Tree
d. Data

44. What does the usage of Preemption and Transaction Rollback prevent?

(a) Deadlock situation

(b) Data manipulation

(c) Unauthorised usage of data files

(d) File preemption

45. Transaction manager_____

a. Maintains before and after database image
b. Maintains appropriate concurrency control
c. Maintains a log of transactions
d. None of these

46. _____ contains complete record of all activities that affected the content of a database during a certain period of time.

a. Report writer
b. Transaction log
c. Data manipulation language
d. Query language

47. _____ is the highest isolation level in transaction management.

a. Repeated read
b. Uncommitted read
c. Committed read
d. Serializable

48. _____ are the commands used to control access over objects in relational database.

a. GRANT & REVOKE
b. QUE & QUIST
c. CASCADE & MVD
d. None of the above

49. In a relation scheme R = (A, B, C, D, E, H) on which the following functional dependencies hold: {A–>B, BC–>D, E–>C, D–>A}, the candidate keys of R are_____

a. AE, BE

b. AE, BE, DE

c. AEH, BEH, BCH

d. AEH, BEH, DEH

50. Select the TRUE statement from the following:

a. A relation R is in 3NF if every non-prime attributes of R is completely functionally dependent on every key of R

b. Every relation in 3NF is also in BCNF

c. No relation can be in both 3NF and BCNF

d. Every relation in BCNF is also in 3NF

1. Cyber Security provide security against what?

1.

A. Against Malware

B. Against cyber-terrorists

C. Defends a device from threat.

D. All mentioned options

Answer: D

2. Who is popularly known as the father of cyber security?

1.

A. Robert

B. August Kerckhoffs

C. Bob Thomas

D. Charles

Answer: B

3. Which of the below is a kind of cyber security?

1.

A. Cloud Security

B. Application Security

C. Cloud Security

D. All options mentioned above

Answer: D

4. Which of the below does not constitute a cybercrime?

1.

A. refusal of service

B. Man in the middle

C. Phishing

D. AES

Answer: D (AES stands for Advanced Encryption Standard) safeguards data by encoding it)

5. Which of the below benefits of cyber security is not true?

1.

A. System getting slower

B. Computer lagging and crashes

C. provide privacy to users

D. Secures system against viruses

Answer: A

6. Who coined the term " Cyberspace"?

1.

A. William Gibson

B. Andrew Tannenbaum

C. Scott Fahlman

D. Richard Stallman

Answer: A

7. Which of the below is a hacking technique in which cybercriminals create fictitious web pages or domains to deceive or obtain more traffic?

1.

A. Pharming

B. Mimicking

C. Spamming

D. Website-Duplication

Answer: A (Pharming is a method and technique used by cybercriminals to deceive people by creating fake websites and domains)

8. Which of the below is a popular victim of cyber attackers looking to gain the IP address of a target or victim user?

1.

A. emails

B. websites

C. IP tracer

D. web pages

Answer: B

9. Which of the below does not qualify as a form of peer-to-peer cybercrime?

1.

A. Trojans are implanted into a targeted device.

B. On the deep web, payment information is leaked.

C. mm

D. Phishing

Answer: B

10. An act to injure, corrupt, or threaten a system or network is characterised as which of the below?

1.

A. Digital crime

B. Threats

C. System hijacking

D. Cyber Attack

Answer: D

11. Any company or agency's IT protection is managed and maintained by?

1.

A. Software Security Specialist

B. CEO of the organisation

C. Security Auditor

D. IT Security Engineer

Answer: D (This is a profession in which a professional develops and maintains various systems and security devices for the firm or organisation)

12. Which of the below is an internet fraud in which a consumer is digitally persuaded to reveal personal data by cybercriminals?

1.

A. MiTM attack

B. Phishing attack

C. Website attack

D. DoS attack

Answer: B

13. Which of the below security encryption standards is the weakest?

1.

A. WPA3

B. WPA2

C. WPA

D. WEP

Answer: D (WEP stands for wired Equivalent Privacy. It is a most insecure encrypted protocol)

14. Which of the below measures can help reduce the risk of data leakage?

1.

A. Steganography

B. Chorography

C. Cryptography

D. Authentication

Answer: A (Steganography functions as concealing information in a non-suspicious manner)

15. This is the concept for guiding information security policy within a corporation, firm, or organisation. What exactly is "this" in this context?

1.

A. Confidentiality

B. Non-repudiation

C. CIA Triad

D. Authenticity

Answer: C (CIA Triad is the most popular and frequently used approach, focusing on the confidentiality of information)

16. _____ means the security of data from tampering by unidentified users.

1.

A. Confidentiality

B. Integrity

C. Authentication

D. Non-repudiation

Answer: B

17. Which of the below implemented is not a good means of safeguarding privacy?

1.

A. Biometric verification

B. ID and password-based verification

C. 2-factor authentication

D. switching off the phone

Answer: D

18. When ____ and ____ are in charge of data, the integrity of the data is imperilled?

1.

A. Access control, file deletion

B. Network, file permission

C. Access control, file permission

D. Network, system

Answer: C

19. The authenticity and security of data travelling over a network are ensured by?

1.

A. Firewall

B. Antivirus

C. Pentesting Tools

D. Network-security protocols

Answer: D (Network-security protocols govern the procedures and processes used to protect network data against illegal content collection)

20. _____ creates an isolated passage across a public network that enables computing devices to communicate and receive data discreetly as though they were directly linked to the private network.

1.

A. Visual Private Network

B. Virtual Protocol Network

C. Virtual Protocol Networking

D. Virtual Private Network

Answer: D

21. _____ is one of the safest Linux operating systems, offering invisibility and an incognito mode to protect user data.

1.

A. Fedora

B. Tails

C. Ubuntu

D. OpenSUSE

Answer: B

22. What is a single entrance for multiple connectivities called?

1.

A. Web services

B. phishing

C. Directory service

D. worms

Answer: C (A directory service is the following system of software and protocols that keep track of knowledge about the company, clients, or sometimes both)

23. Circuit-level gateway firewalls have which of the below disadvantages?

1.

A. They're expensive

B. They're complex in architecture

C. They do not filter individual packets

D. They're complex to setup

Answer: C

24. The initial phase of ethical hacking is?

1.

A. DNS poisoning

B. Footprinting

C. ARP-poisoning

D. Enumeration

Answer: B (In this Phase, The attacker attempts to find as many attack vectors as he can, reconnaissance is another term for footprinting)

25. Which of the below can be classified as a type of computer threat?

1.

A. Dos Attack

B. Phishing

C. Soliciting

D. Both A and C

Answer: A (A denial of service attack is referred to as a dos attack, it's a type of cyber-attack in which someone tries to prevent a machine from serving its intended consumers)

26. In system hacking, which of the below is the most crucial activity?

1.

A. Information gathering

B. Covering tracks

C. Cracking passwords

D. None of the above

Answer: C

27. When the number of users surpasses the network's capacity, which of the below network factors suffers the most?

1.

A. Reliability

B. Performance

C. Security

D. Longevity

Answer: D

28. Which of the below cyber security principles states that the security system should be as compact and straightforward as possible?

1.

A. Open-design

B. The economy of the Mechanism

C. Least privilege

D. Fail-safe Defaults

Answer: B

29. Which of the below malware types permits the hackers to access administrative controls and do nearly everything he wants with the infected systems?

1.

A. RATs

B. Worms

C. Rootkits

D. Botnets

Answer: A (RATs stands for Remote Access Trojans which gives the attacker administrative power over your device, just as if they had physical access)

30. The first hacker's conference was held in which of the below locations?

1.

A. OSCON

B. DEVON

C. DEFCON

D. SECTION

Answer: C (DEFCON is one of the most well-known and biggest hackers and security consultant conferences in the world. It's often conducted in Las Vegas, Nevada, once a year)

31. When any IT device, service, or system requires security checks, the term "security testing" is employed.?

1.

A. Threat

B. Vulnerability

C. Objective of evaluation

D. Attack

Answer: C (When any IT system, device, or platform requires assessment for safety purposes or to address any faults after being evaluated by security researchers, the term "objective of evaluation" is used)

32. Which of the below is used to analyse network flow and monitor traffic?

1.

A. Managed detection and response

B. Cloud access security broker

C. Network traffic analysis

D. Network traffic analysis

Answer: C

33. Which of the below is a method of gaining access to a computer program or an entire computer system while circumventing all security measures?

1.

A. Backdoor

B. Masquerading

C. Phishing

D. Trojan Horse

Answer: Backdoor (A backdoor is a sort of malware that bypasses standard authentication mechanisms to gain access to a system)

34. The term "protection from _____of source code" refers to limiting access to the source code to just authorised individuals.

1.

A. disclosure

B. alteration

C. destruction

D. log of changes

Answer: C

35. _____ are programmes or procedures that enable hackers to maintain control of a computer system.?

1.

A. Exploits

B. Antivirus

C. Firewall by-passers

D. Worms

Answer: A (Exploits are programs or algorithms that allow hackers to gain total control of a computer system)

1) What is the protocol data unit (PDU) for the application layer in the Internet stack?

a. Frame

b. Datagram

c. Segment

d. Message

2) Which among this is not a client-server application?

a. Ping

b. Web-browsing

c. Internet Chat

d. Email

3) If an Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it:245.248.128.0/20 and the ISP wants to give half of this chunk of address to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Then, which among the following is a valid allocation of addresses to A and B?

a. 245.248.132.0/22 and 245.248.132.0/21

b. 245.248.136.0/21 and 245.248.128.0/22

c. 245.248.128.0/21 and 245.248.128.0/22

d. 245.248.136.0/22 and 245.248.132.0/21

4) The pairs of OSI protocol layer/sub-layer and its functionality are given below, the INCORRECT pair from among these are:

a. Data Link Layer and Bit synchronization

b. Network Layer and Routing

c. Medium Access Control sub-layer and Channel sharing

d. Transport Layer and End-to-end process communication

5) A firewall is installed at the point where the secure internal network and untrusted external network meet, which is also known as_____

a. Secure point

b. Meeting point

c. Chock point

d. Firewall point

6) What does DHCP stand for?

a. Dynamic Host Configuration Protocol

b. Dynamic Host Configuration Provider

c. Digital Host Communication Provider

d. Digital Host Communication Protocol

7) The growth of congestion window occurs_____

a. Infinitely

b. Up to timeout

c. Up to threshold

d. Up to the size of receiver's window

8) When was Aloha developed?

a. In 1990

b. In 1980

c. In 1970

d. In 1960

9) Which of these is a Transmission media that can be used in LAN?

a. fibre optics

b. coaxial cable

c. microwave

d. satellite

10) Which of these protocols given below uses UDP as transport protocol?

a. DNS

b. Telnet

c. HTTP

d. SMTP

11) LANs can be connected by a device known as _____

a. Ethernet card

b. Modems

c. Routers

d. Bridges

12) What is the bit rate In Ethernet when Manchester encoding is used?

a. Same as the baud rate

b. Twice the baud rate

c. Half the baud rate

d. None of these

13) What is slot time for gigabit Ethernet?

a. 0.612

b. 0.512

c. 0.412

d. 0.312

14) Radio waves are _____

a. directional

b. bidirectional

c. omnidirectional

d. unidirectional

15) The term that refers to the shortest unit and carries data elements is_____

a. Single element

b. Data carrier

c. Signal element

d. Both b and c

16) Which architecture is FTP built on?

a. P2P

b. Client-server

c. Both the above

d. None of these

17) What is the address resolution protocol (ARP) used for?

a. Finding the IP address of the default gateway

b. Finding the MAC address that corresponds to an IP address

c. Finding the IP address that corresponds to a MAC address

d. Finding the IP address from the DNS

18) Which of the following are not true characteristics of NetBEUI?

a. Fast for small networks to Self-tuning

b. Little configuration required

c. Routable

d. Highly customizable

19) Identify the correct option for the given assertion and reason

Assertion: In IEEE 802.3 is not suitable for client server application

Reason: In Ethernet, there is a restriction on minimum size.

a. Assertion and Reason, both are true

b. Assertion and reason are false

c. Assertion is true and reason is false

d. Assertion and reason are true, but this is not the correct reason for the given assertion

20) Consider that a computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps?

a. 8 seconds

b. 5 seconds

c. 2 seconds

d. 1.6 seconds

21) In Unicast Routing, we use shortest path tree method for finding shortest path of the_____

a. Nodes

b. Networks

c. Packets

d. Frames

22) Which is the network that provides high speed connectivity?

a. Internetwork

b. WAN

c. LAN

d. MAN

23) What is the maximum number of subnets and the maximum number of hosts in each subnet, if the address of a class B host is to be split into subnets with a 6-bit subnet number?

a. 64 subnets and 262142 hosts

b.62 subnets and 262142 hosts

c. 64 subnets and 1024 hosts

d. 62 subnets and 1022 hosts

24) Which of the following statements about HTTP are false?

a. HTTP describe the structure of web page

b. HTTP run over TCP

c. HTTP can be used to test the validity of hyper link test

d. HTTP allows information to be stored in url

25) What is an anticipated result from multi programming operations?

a. Handling of more jobs

b. Reduced computer idle time

c. Better scheduling of work

d. All of the above

26) What is the Network, in which every computer is capable of playing the role of client, server or both at the same time?

a. dedicated server network

b. local area network

c. wide area network

d. peer-to-peer network

27) Match the following:

(P) SMTP	**(1) Application layer**
(Q) BGP	**(2) Transport layer**
(R) TCP	**(3) Data link layer**
(S) PPP	**(4) Network layer**
	(5) Physical layer

(a) P – 2 Q – 1 R – 3 S – 5

(b) P – 1 Q – 4 R – 2 S – 5

(c) P – 1 Q – 4 R – 2 S – 3

(d) P – 2 Q – 4 R – 1 S – 3

28) If a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits second) over a 1 km(kilometer) cable with no repeaters and the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

a. 16000

b. 20000

c. 10000

d. 8000

29) One of the header fields in an IP datagram is the Time to Live (TTL) field. Which of these below given statements best explains the need for this field?

(a) It can be used to reduce delays

(b) It can be used to optimize throughput

(c) It can be used to prioritize packets

(d) It can be used to prevent packet looping

30) A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first back off race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second back off race is_____

a. 0.625

b. 0.75

c. 0.5

d. 1.0

31) In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then what is the optimum packet size?

a. 4

b. 6

c. 7

d. 9

32) Station B needs to send a message consisting of 9 packets to Station C using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that B transmits gets lost (but no acks from C ever get lost), then what is the number of packets that B will transmit for sending the message to C?

a. 14

b. 16

c. 12

d. 18

33) Which of the following assertions about the Internet Protocol (IP) is False?

a. IP Packets from the same source to the same destination can take different routes in the network

b. The packet source cannot set the route of an ongoing packets; the route is only determined by the routing tables in the routers on the way

c. IP ensures that a packet is discarded if it is unable to reach its destination within a given number of hops

d. It is possible for a computer to have multiple IP addresses

34) For an organization that has a class B network and wishes to form subnets for 64 departments, what will the subnet mask be?

a. 255.255.64.0

b. 255.255.64.0

c. 255.255.128.0

d. 255.255.252.0

35) The message 11001001 is to be transmitted using the CRC polynomial x^3 + 1 to protect it from errors. The message that should be transmitted is_____

a. 11001001011

b. 11001001000

c. 110010010011

d. 11001010

36) If two computers C_1 and C_2 are configured as follows and C_1 has IP address 203.197.2.53 and netmask 255.255.128.0, while C_2 has IP address 203.197.75.201 and netmask 255.255.192.0, then which of the following statements is true?

a. C_2 assumes C_1 is on the same network, but C_1 assumes C_2 is on a different network

b. C_1 assumes C_2 is on the same network, but C_2 assumes C_1 is on a different network

c. C_1 and C_2 both assume they are on different networks

d. C_1 and C_2 both assume they are on the same networks

37) Look at these three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links

- **[S_1] The computational overhead in link state protocols is higher than in distance vector protocols.**
- **[S_2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.**
- **[S_3] After a topology change, a link state protocol will converge faster than a distance vector protocol.**

Which one of the following is correct about S_1, S_2, and S_3?

a. S_1, S_2, and S_3 are all false

b. S_1, S_2, and S_3 are all true

c. S_1 and S_2 are true, but S_3 is false

d. S_1 and S_3 are true, but S_2 is false

38) Which of these functionalities should be implemented by a transport protocol over and above the network protocol?

a. Detection of duplicate packets

b. End to end connectivity

c. Packet delivery in the correct order

d. Recovery from packet losses

39) If the length of a Broadcast LAN is 2 Km and it has 107 bps bandwidth and uses CSMA/CD and the signal travels along the wire at 2108m/s, then what is the minimum size of the packet?

a. 100 bytes

b. 200 bytes

c. 50 bytes

d. None of these

40) Which is one of the pairs of protocols given below from both the protocols that can use multiple TCP connections between the same client and the server?

a. HTTP, TELNET

b. HTTP, SMTP

c. FTP, SMTP

d. HTTP, FTP

41) What is the number of 8-bit characters that can be transmitted per second over a 9600 baud serial communication link that use asynchronous mode of transmission with 8 data bits, one start bit and one parity bit?

a. 800

b. 600

c. 876

d. 1200

42) In the starting phase of TCP congestion control algorithm size of congestion control window_____

a. Increase Linearly

b. Does not increase

c. Increase exponentially

d. Increase

43) If a sender is employing public key Cryptography to send a secret message to a receiver, which one of the following statements is true?

a. Sender encrypts using his own public key

b. Receiver decrypts using his own public key

c. Sender encrypts using receiver's public key

d. Receiver decrypts using sender's public key

44) If everyone in a group of N people want to communicate secretly with the N – 1 others using symmetric key cryptographic system and the communication between any two persons should not be decodable by the others in the group, then what is the number of keys required in the system as a whole to satisfy the confidentiality requirement?

a. N(N-1)

b. N(N-1) / 2

c. 2N

d. $(N-1)^2$

45) Internetwork is made of five networks. Which are those?

a. Four LANs and One MAN

b. Four LANs and One WAN

c. Three LANs and Two WAN

d. Four MANs and One WAN

46) This is one of two protocols that let a network administrator supervise and distribute IP addresses from a central point. It automatically sends a new IP address when a computer is plugged into a different place in the network.

a. UDP

b. TCP

c. TCPMAN

d. DHCP

47) A sender B sends a message m to receiver A, which is digitally signed by B with its private key. In this scenario, one or more of the following security violations can take place.

I. B can launch a birthday attack to replace m with a fraudulent message

II. A third party attacker can launch a birthday attack to replace m with a fraudulent message

III. A can launch a birthday attack to replace m with a fraudulent message

Which of the following are possible security violations?

a. I only

b. II only

c. II and III only

d. I and II only

48) Take the example of 1 Mbps error-free line, with the maximum frame size of 1000 bits and new packets generated about 1 second apart . Timeout interval is 10 m/sec if the special acknowledgement timer were eliminated, unnecessary timeout would occur . In the scenario, how any times average msg would have been transmitted?

a. Twice

b. Only once

c. Thrice

d. None of the above

49) _____ has the responsibility for Accounting functions.

a. Transport layer

b. Data link layer

c. Data layer

d. Network layer

50) In a class C address, if 3 bits are borrowed for subnetting of host ID part, the total possible number of subnet masks are _____.